

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	Analog IC Design Lab
Subject Code:	EE 653
Assignment No:	07
Submission Date:	18 March 2025
Instructor Name:	Professor Madhav Pathak

TASK 1: Designing a Low Pass Filter using Ideal OpAmp

- (Refer to Fig. 1) Design a second-order unity-gain active low-pass filter with a cutoff frequency of 50 kHz using the Sallen-Key topology. Plot and compare the frequency response of the Butterworth, Bessel, and Chebyshev filters. Also plot the time-domain response in the passband and stopband.

(In this task, the design should be carried out using an Ideal OpAmp.)

Also report the roll-off (in dB) for all three filters.

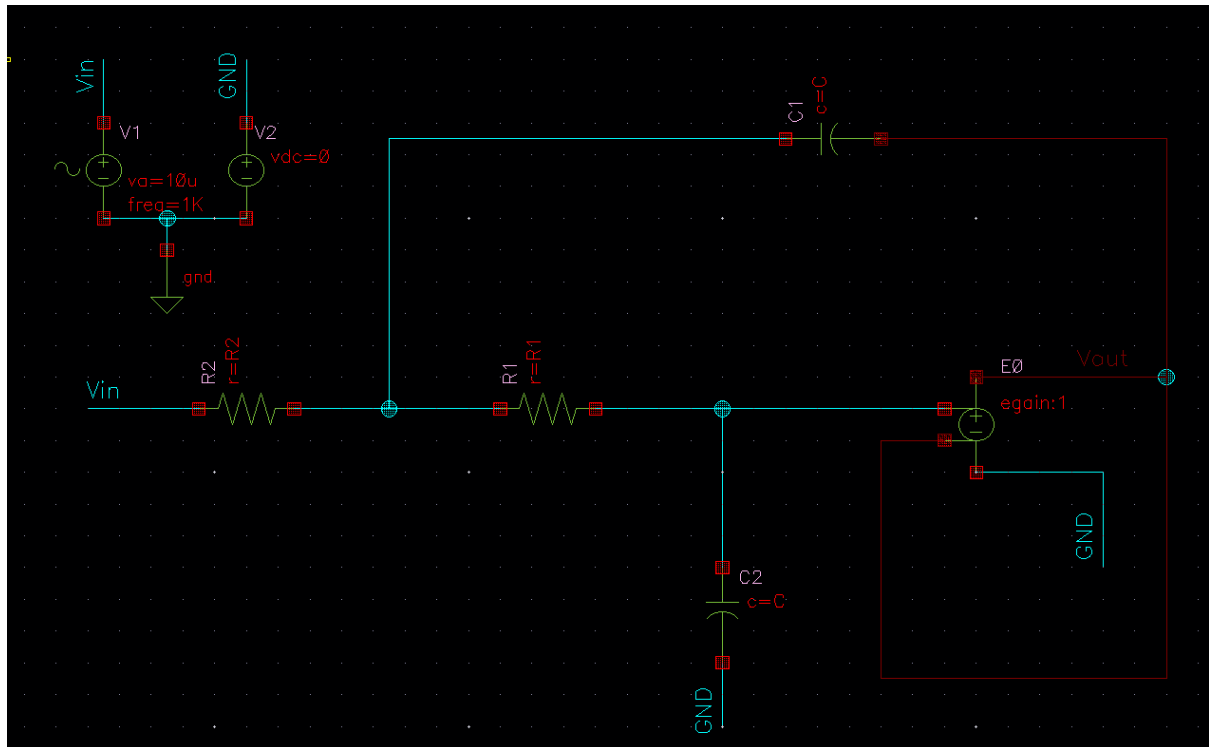


Figure 1: Schematic

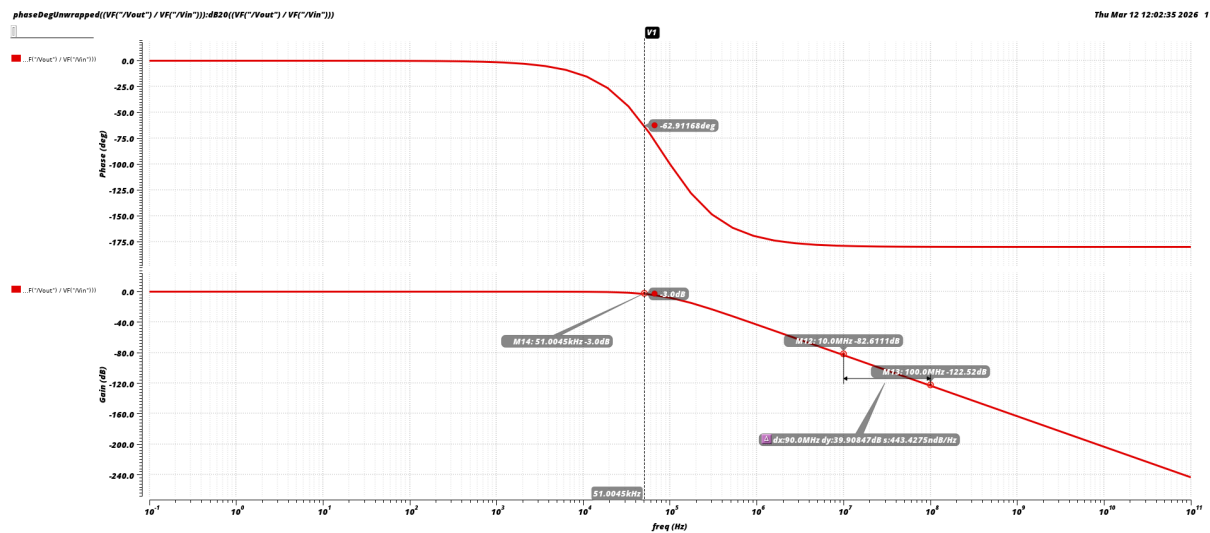


Figure 2: Butterworth

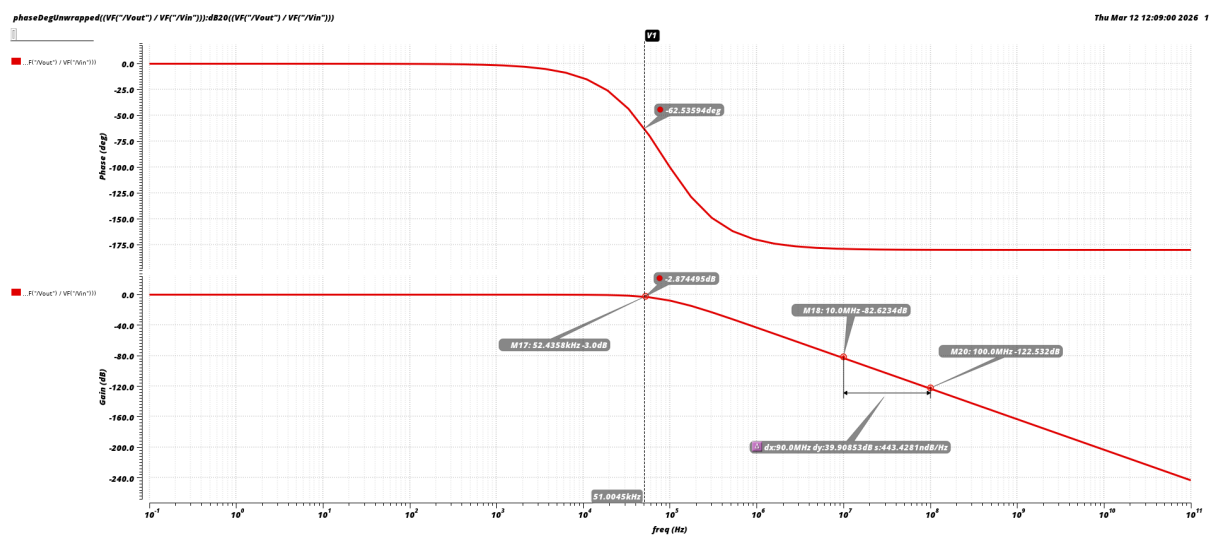


Figure 3: Bessel

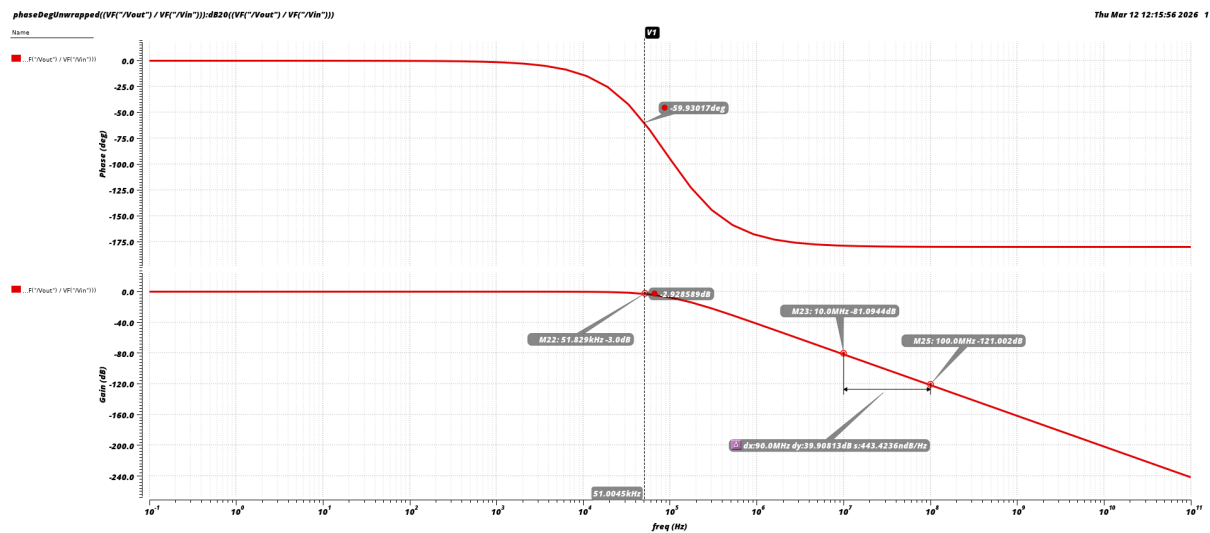


Figure 4: Chebyshe

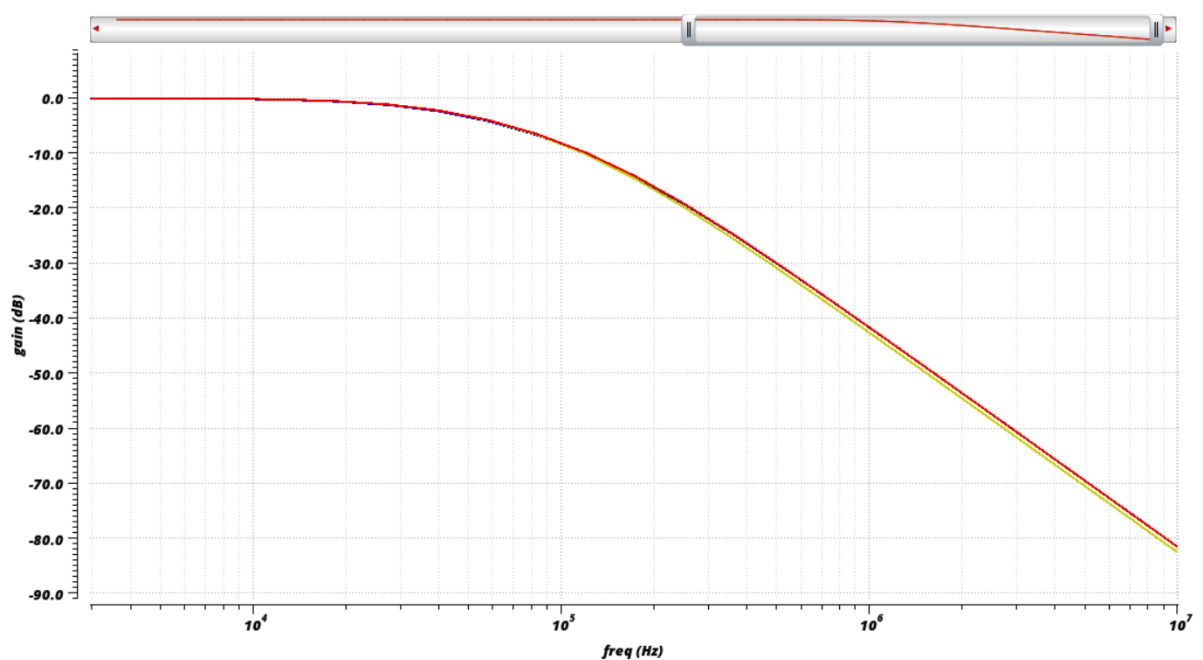


Figure 5: Roll off comparison

TASK 2: Designing a Low Pass Filter using Designed OpAmp [5pts]

- (Refer to Fig. 1) Design a second-order unity-gain active low-pass filter with a cutoff frequency of 50 kHz, using Sallen-Key Topology. Plot and compare the frequency response of the Butterworth, Bessel, and Chebyshev filters. Plot the time domain response in the passband and stopband. (In this task, you need to do the design using your designed OpAmp.) Also, report the Roll-Off (dB) for all three Filters.

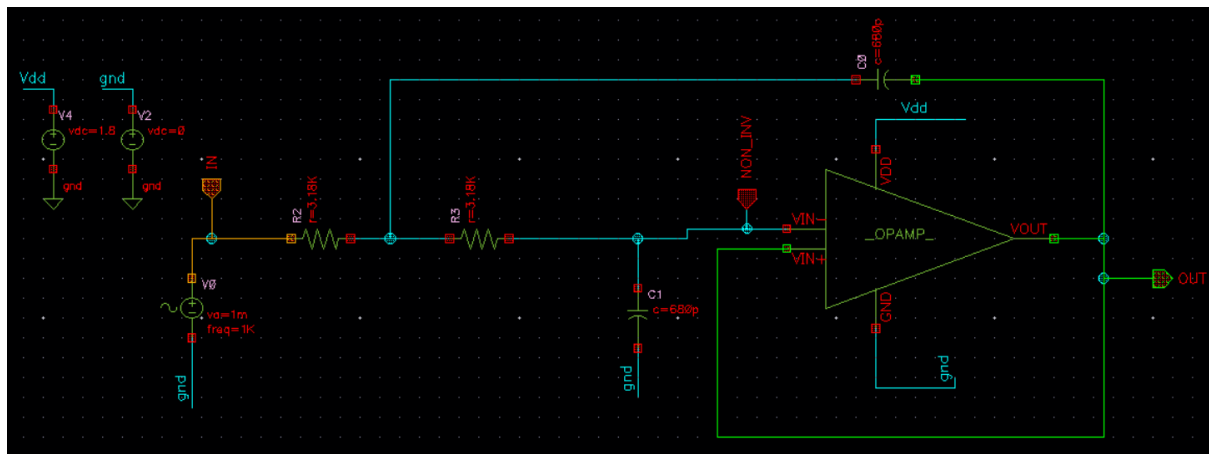
BUTTERWORTH:

Figure 6: Schematic

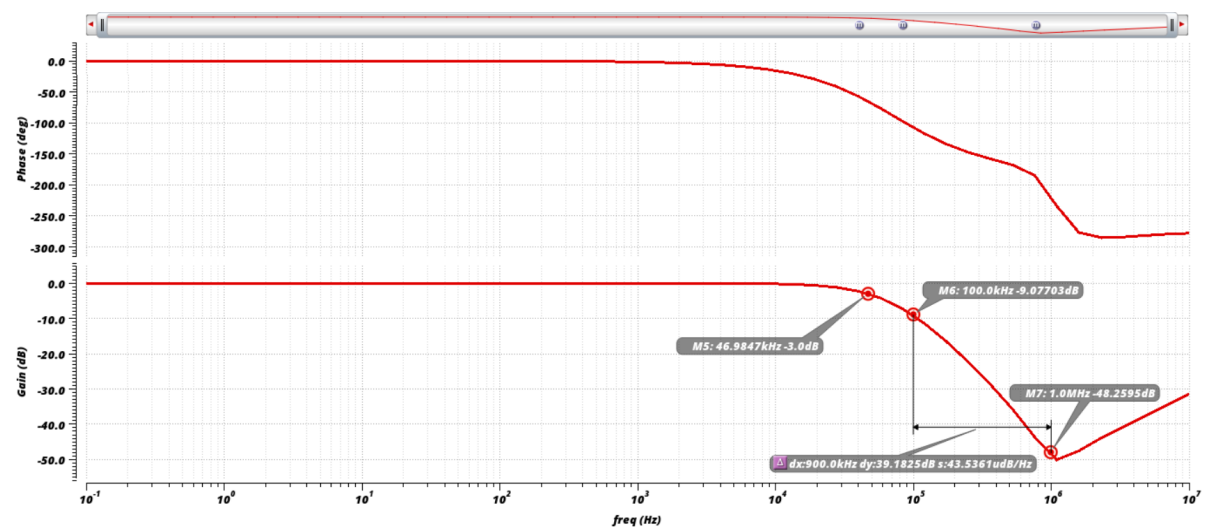


Figure 7: Butterworth

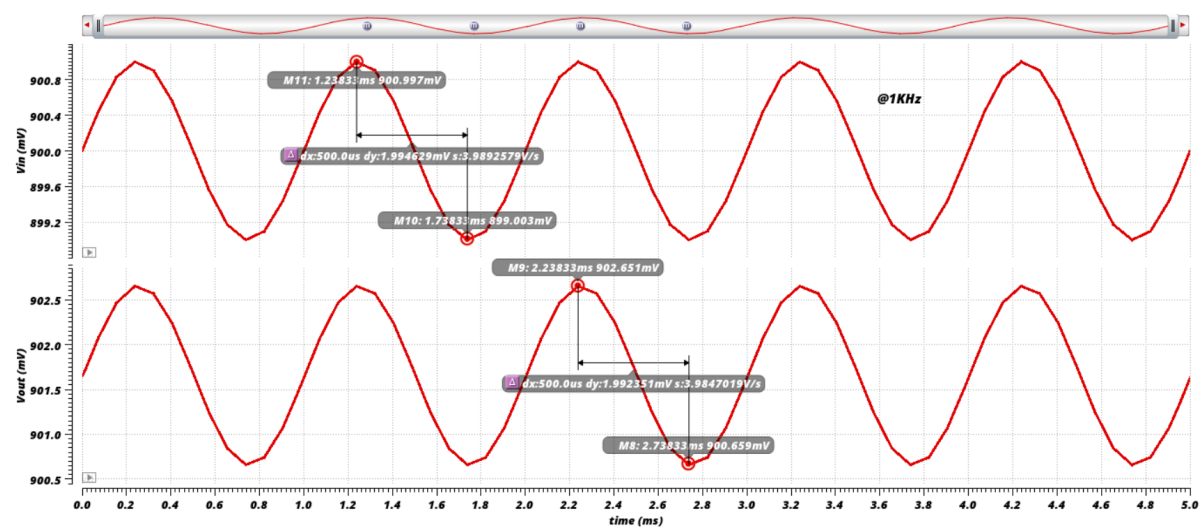


Figure 8: Butterworth Pass Band

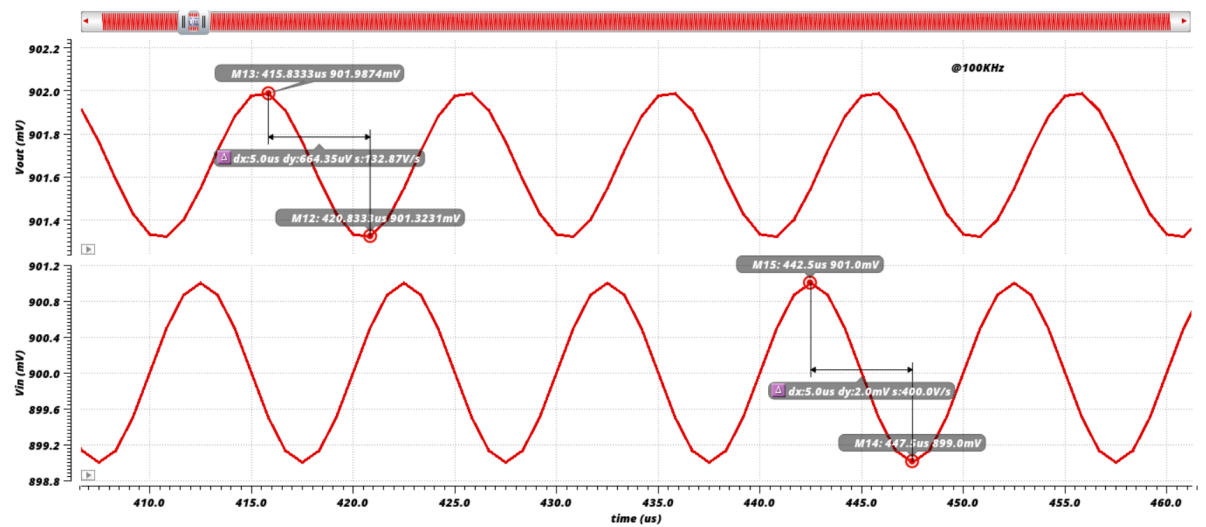


Figure 9: Butterworth Stop Band

Component	Value
R_1	3.18 k Ω
R_2	3.18 k Ω
C_1	680 pF
C_2	680 pF

Table 1: Component Values for Sallen-Key Low Pass Filter (BUTTERWORTH)

BESSEL:

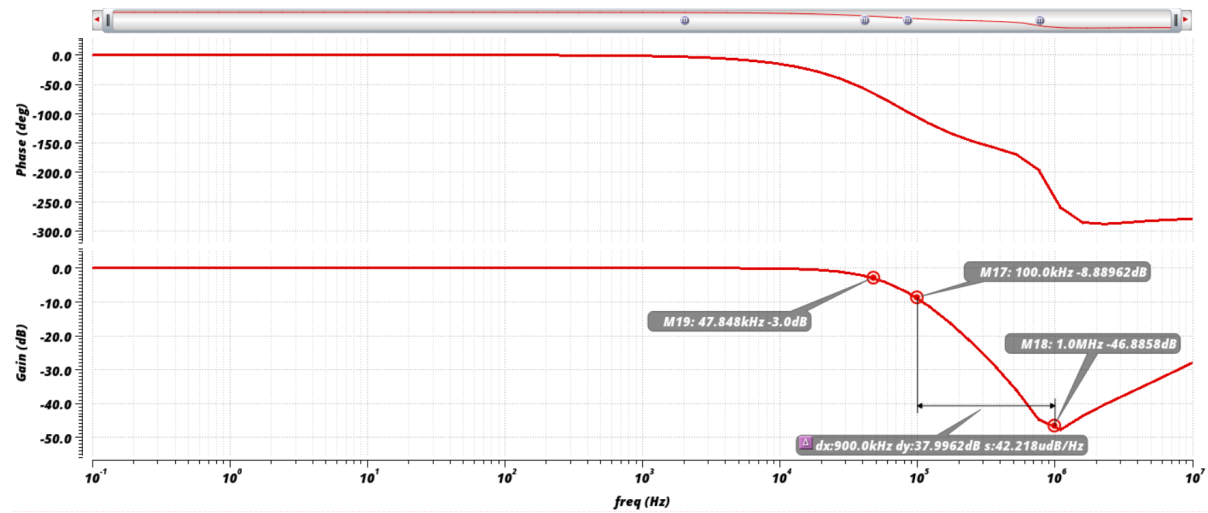


Figure 10: Bessel

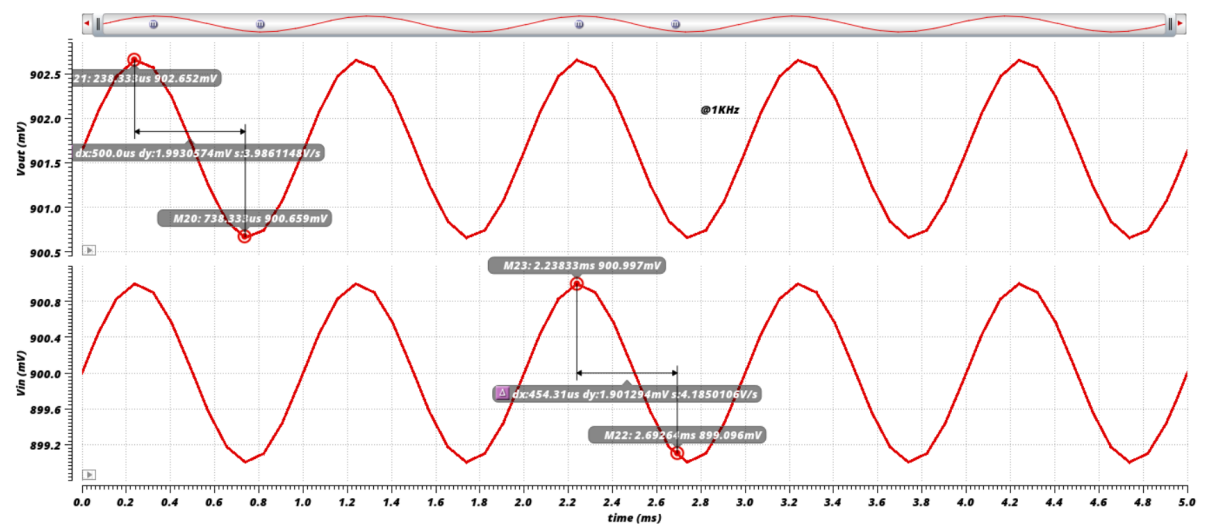


Figure 11: Bessel Pass Band

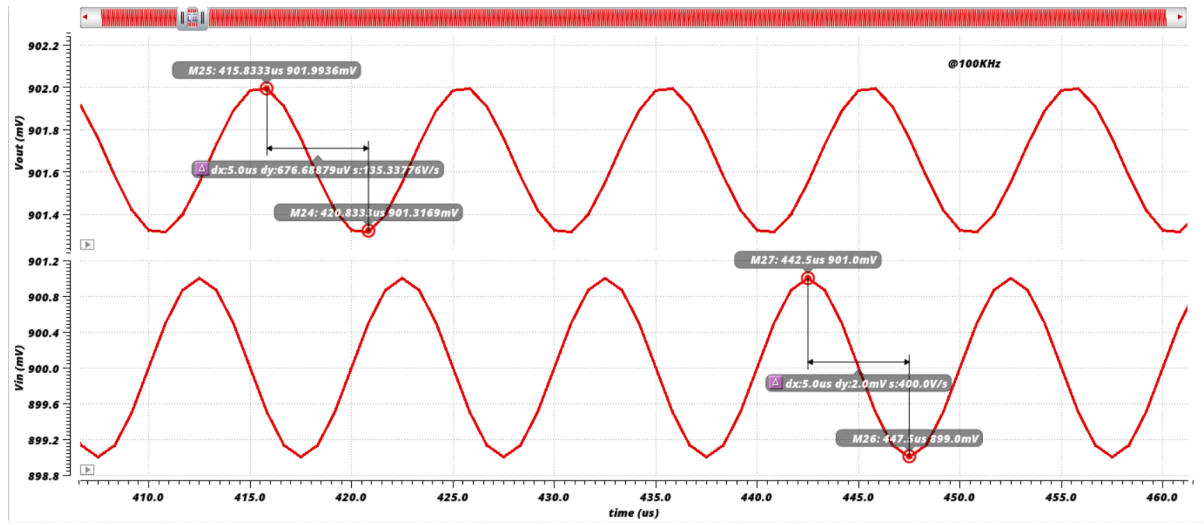


Figure 12: Bessel Stop Band

Component	Value
R_1	2.12 k Ω
R_2	2.12 k Ω
C_1	1 nF
C_2	1 nF

Table 2: Component Values for Sallen-Key Low Pass Filter (BESSEL)

Chebyshev:

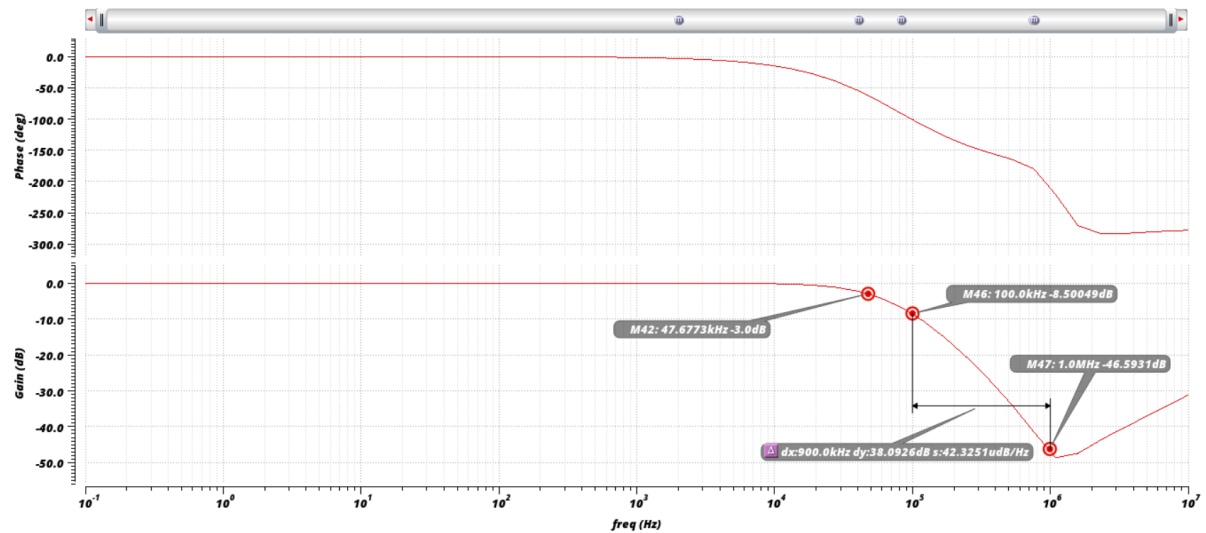


Figure 13: Chebyshev

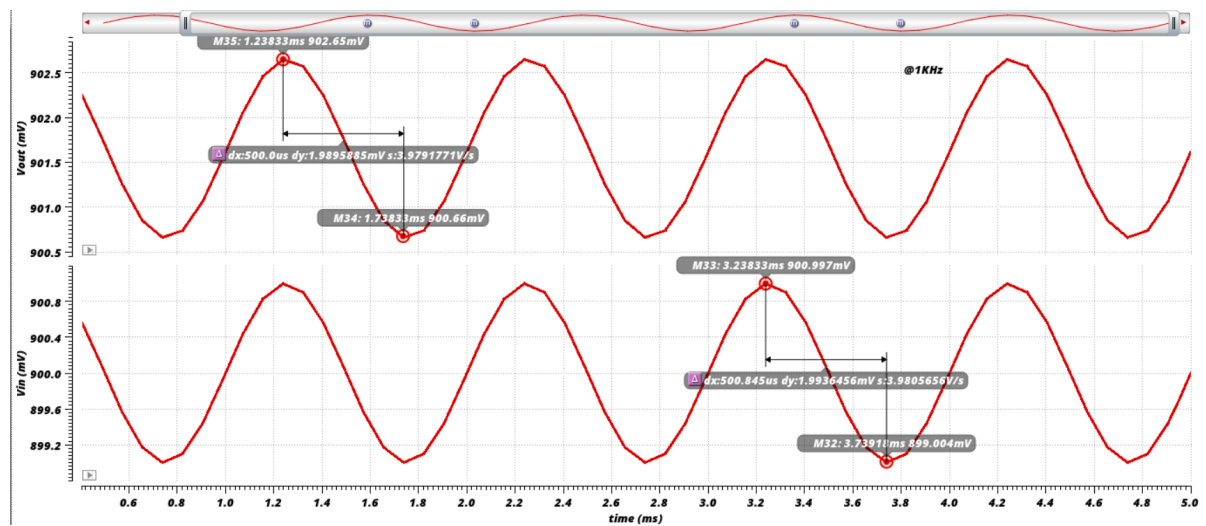


Figure 14: Chebyshev Pass Band

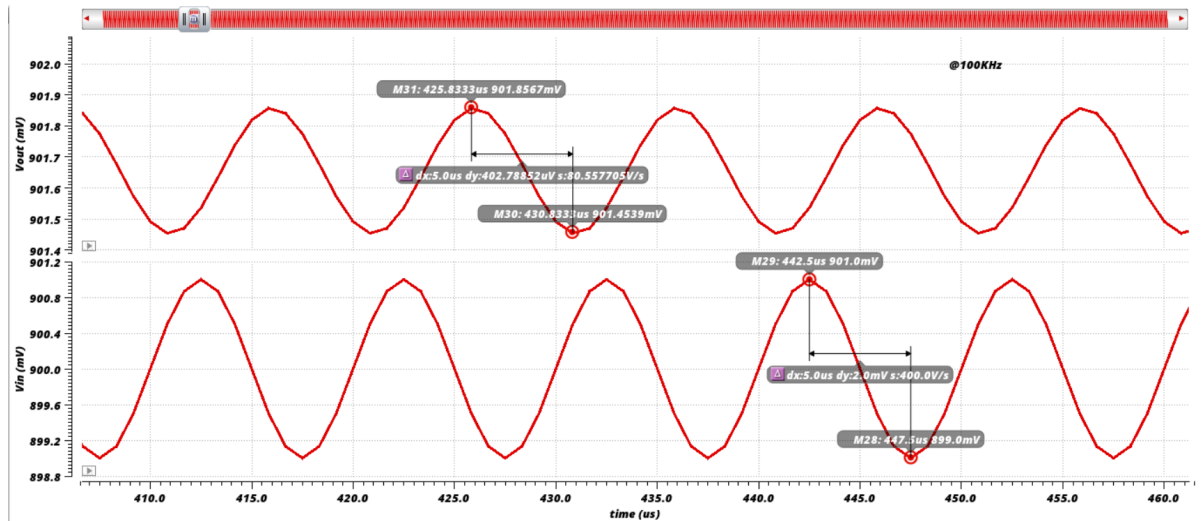


Figure 15: Chebyshev Stop Band

Component	Value
R_1	3.58 k Ω
R_2	3.58 k Ω
C_1	680 pF
C_2	680 pF

Table 3: Component Values for Sallen-Key Low Pass Filter

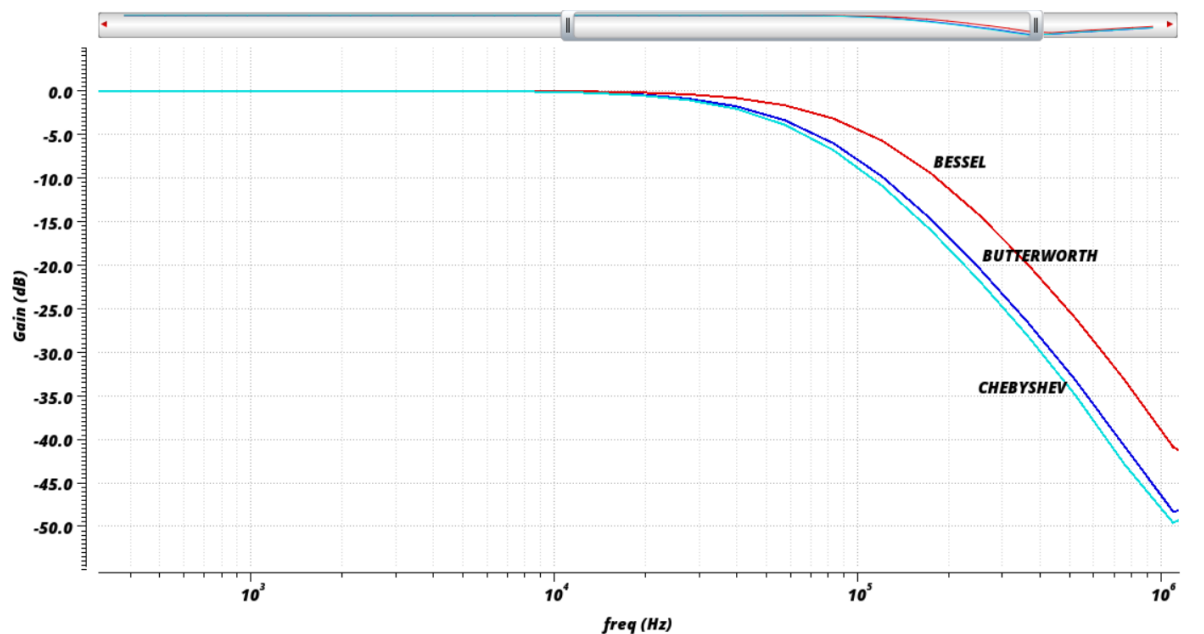


Figure 16: comparison of roll-off

TASK 3: Designing an Anti-Aliasing Filter using Designed OpAmp [3pts]

- Design an antialiasing filter with a cutoff frequency of 10 KHz for a system requiring more than 72dB stop-band attenuation, i.e., at the Nyquist frequency of 60 KHz. Provide the necessary plots in the time and frequency domains. (Frequency Response, Time Domain Response in Passband and StopBand). (In this task, you need to do the design using your designed OpAmp.)

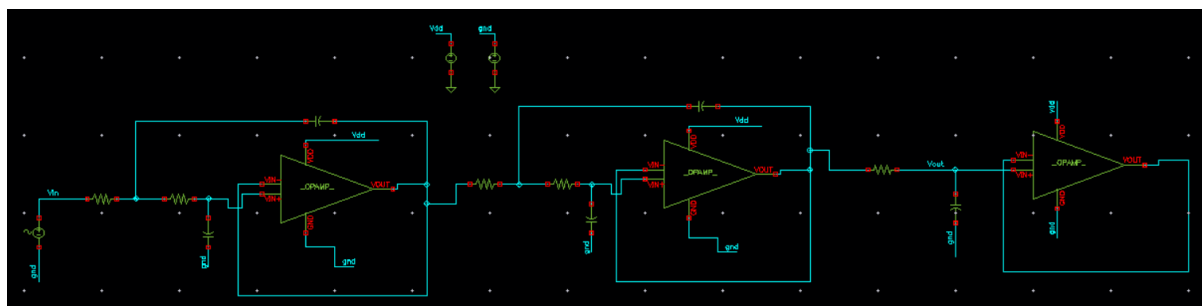


Figure 17: Chebyshev

calculation of order (n)

Given:

- Cutoff frequency: $f_c = 10, \text{ kHz}$
- Stopband frequency: $f_s = 60, \text{ kHz}$
- Required attenuation: $A \geq 72, \text{ dB}$

For an n^{th} order low-pass filter:

$$A = 20n \log_{10} \left(\frac{f_s}{f_c} \right) \quad (1)$$

Substituting values:

$$72 \leq 20n \log_{10} \left(\frac{60}{10} \right) \quad (2)$$

$$72 \leq 20n \log_{10}(6) \quad (3)$$

$$\log_{10}(6) \approx 0.778 \quad (4)$$

$$72 \leq 20n \times 0.778 \quad (5)$$

$$72 \leq 15.56n \quad (6)$$

$$n \geq \frac{72}{15.56} \quad (7)$$

$$n \geq 4.63 \quad (8)$$

$$\therefore n = 5 \quad (9)$$